NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2024-2025

EE7204 – LINEAR SYSTEMS

November / December 2024 Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 5 questions and comprises 5 pages.
2. Answer all 5 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.
6. Laplace–Transform Table is included in Appendix A on page 5.

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1. (a) Consider the linear time-invariant system

$$\dot{x}(t) = \begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t), \quad x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$
$$y(t) = \begin{bmatrix} 1 & 1 \end{bmatrix} x(t),$$

where x is the state, y is the output and u is the control input. Find the output response $y(t)$ when the input is given by $u(t) = e^{-t}, t \geq 0$.

(8 Marks)

(b) Consider the linear time-invariant system

$$\dot{x}(t) = Ax(t), x(0) = x_0$$

where x is the state and

Note: Question No. 1 continues on page 2.

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$$A = \begin{bmatrix} -1 & 0 & 1 \\ 1 & 0 & -1 \\ 1 & 1 & -1 \end{bmatrix}.$$

(i) Find a linear transformation that transforms the matrix into the diagonal or Jordan form and thus compute e^{At}.

(ii) Is the system stable in the sense of Lyapunov? Furthermore, is the system asymptotically stable?

(12 Marks)

2. Consider the linear time-invariant system

$$\dot{x}(t) = Ax(t) + Bu(t),$$
$$y(t) = Cx(t),$$

where x is the state, u is the input, y is the output,

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 1 \\ -1 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

(a) Determine if the system is controllable and observable.

(8 Marks)

(b) Find the canonical decomposition of the system and determine if the system is detectable.

(8 Marks)

(c) Find the transfer function of the system.

(4 Marks)

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3. (a) Consider the nonlinear system

$$\dot{x}_1 = x_2 - x_1 x_2^2,$$
$$\dot{x}_2 = -2x_1 - x_2^3$$

Find the equilibrium state of the system. Furthermore, determine if the equilibrium state is globally asymptotically stable by choosing Lyapunov function candidate $V(x) = x_1^2 + \frac{1}{2}x_2^2$.

(10 Marks)

(b) Given a polynomial matrix

$$P(s) = \begin{bmatrix} s+2 & s+2 \\ s^2+5s+6 & s^2-4 \end{bmatrix}$$

determine whether there exists a unimodular matrix $U(s)$ such that $U(s)P(s)$ is a diagonal polynomial matrix.

(10 Marks)

4. Given a strictly proper rational transfer function $G(s)$, answer the following questions.

$$G(s) = \begin{bmatrix} \frac{s}{(s+2)(s+5)} & \frac{1}{(s+1)(s+2)} & \frac{1}{s} \\ \frac{1}{s+2} & \frac{2}{(s+1)(s+2)} & \frac{1}{s+5} \end{bmatrix}$$

(a) Determine a Right Polynomial Fraction Description (RPFD) of $G(s)$, where $D(s)$ is specifically chosen as

$$D(s) = \begin{bmatrix} (s+2)(s+5) & 0 & 0 \\ 0 & (s+1)(s+2) & 0 \\ s^2(s+5) & 0 & s(s+5) \end{bmatrix}.$$

(5 Marks)

Note: Question No. 4 continues on page 4.

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(b) Determine whether the calculated RPFD in (a) is a coprime RPFD (CRFPD). If not, find a CRPFD of $G(s)$.

(10 Marks)

(c) Determine whether the CRPFD obtained in (b) is column reduced. If not, find a column reduced CRPFD of $G(s)$.

(5 Marks)

5. Given the following block diagram in Figure 1, where $G(s) = \frac{1}{s-1}$ is the transfer function of the plant, w and r are step functions, representing external disturbance and reference signal, respectively, answer the following questions:

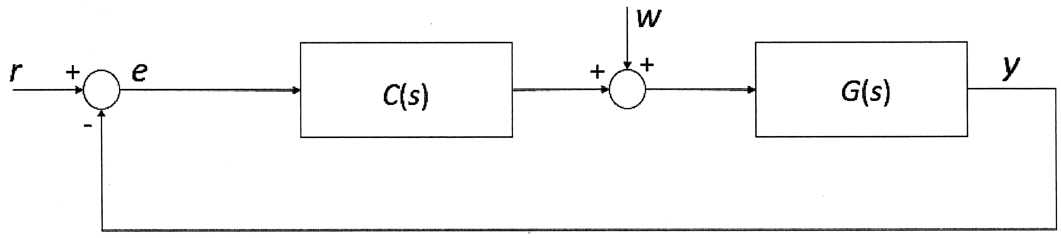

Figure 1

(a) Design a compensator $C(s) = \frac{B(s)}{A(s)}$ such that the transfer function of the closed-loop system has three stable poles at -1, -1, -2, and the system can track any step reference signal asymptotically and reject the impact of $w(t)$ asymptotically.

(12 Marks)

(b) Is it possible to lower the degree of $A(s)$ even further, when designing a compensator $C(s) = \frac{B(s)}{A(s)}$, such that the closed-loop system is stable and can robustly track any step reference signal asymptotically and reject the impact of $w(t)$ asymptotically? If yes, provide one design. If not, give a reason.

(8 Marks)

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APPENDIX A

Laplace-Transform Table

LAPLACE TRANSOFRM $F(s)$	TIME FUNCTION $f(t)$
$\frac{1}{s}$	Unit-step function $u(t)$
$\frac{1}{s+\alpha}$	$e^{-\alpha t}$
$\frac{\alpha}{s(s+\alpha)}$	$1-e^{-\alpha t}$
$\frac{1}{(s+\alpha)^2}$	$te^{-\alpha t}$
$\frac{a}{s^2+a^2}$	$\sin at$
$\frac{s}{s^2+a^2}$	$\cos at$
$\frac{a}{(s+b)^2+a^2}$	$e^{-bt}\sin at$
$\frac{s+b}{(s+b)^2+a^2}$	$e^{-bt}\cos at$
$\frac{\omega_n^2}{s(s^2+2\zeta\omega_n s+\omega_n^2)}$	$1-\frac{1}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_n t}\sin(\omega_n\sqrt{1-\zeta^2}\,t+\theta)$ where $\theta=\cos^{-1}\zeta \qquad (\zeta<1)$
$\frac{\alpha^2}{s(s+\alpha)^2}$	$1-e^{-\alpha t}(1+\alpha t)$

END OF PAPER

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1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.

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